

Dynamic interaction between robot and UAV in aerial manipulation

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Abstract—Aerial manipulation is an emerging field of research, and important applications can be envisaged, such as inspection and maintenance, search and rescue, structure assembly, and logistics. The manipulator transfers forces and torques to the UAV, which affect the UAV position and attitude. This may cause failure or loss of precision in pick and place and assembly operations. In this paper, first, the effect of simple impulsive force and torque disturbances on UAV dynamics is studied. Then, a coupled dynamic model of the UAV and a 1-DOF (Degree of Freedom) manipulator is developed and compared to a simplified decoupled model. Finally, a decoupled dynamic model is proposed for a 3-DOFs manipulator, and the simulation results for a real pick and place operation are presented and discussed. It is evidenced that, in all of the considered scenarios, a lateral displacement of the system is generated during the manipulation in hovering flight, which could jeopardize the manipulator precision.

Keywords—UAV, aerial manipulation, robot, dynamic disturbances, coupled model

I. INTRODUCTION

Aerial manipulation is a new and emerging field of research [1-3]. EU considers it a strategic research line and has recently funded several collaborative projects in this area under the programmes FP7 and H2020. Indeed, aerial manipulators have many important potential application scenarios, such as (i) inspection and maintenance (e.g., in the offshore or nuclear industry), (ii) search and rescue, in case of a disaster or in hazardous environments, (iii) structure assembly (one or more cooperative aerial manipulators), and (iv) logistics (e.g., in an industrial environment).

One of the main issues [1-3] that prevents the effective deployment and diffusion of this technology is that aerial manipulators do not have a stable base and, therefore, forces and torques generated by the manipulator affect the UAV position, attitude, stability, and flight autonomy (since the compensation through attitude control wastes energy). As a consequence of the coupled kinematics and dynamics of UAV and manipulator, the precise positioning of the end-effector of the manipulator for grasping and manipulation is very challenging.

In order to reduce the destabilizing effect and improve the robotic grasping precision, several attempts have been made in the literature: (i) static Center of Gravity (COG) balancing through arm design, (ii) static (COG) balancing through tailored control, (iii) UAV attitude variation minimization through a dedicated control. In particular, Bellicoso et al. [4] proposed a manipulator design that minimizes the variation of COG of the robotic arm during its motion using a static balancing approach. Other authors proposed to limit the arm COG displacement [5] or the attitude variation [6] using redundant arms and a tailored

control. Nevertheless, in these works, the experimental results still show significant COG variations and attitude destabilization during the manipulator motion.

The minimization of the dynamic disturbances transferred by a manipulator to a moving base has been investigated in space robotics [7, 8]. Moreover, in [9], a global optimal method suitable for the minimization of reactions transferred to the manipulator base has been proposed. Nevertheless, the application of these methods to aerial manipulators should be investigated.

Another important challenge in aerial manipulation is related to the vibrations induced on the manipulator and its end-effector by the UAV, which significantly affect the grasping and manipulation precision/performance. The vibration control of an aerial manipulator is still under development. Nevertheless, methods have been already developed for the identification of joint stiffness of a manipulator using Experimental Modal Analysis EMA [10, 11]. Moreover, in [12], the vibration analysis of a heavy payload octocopter has been carried out using EMA, in order to identify a simple Mass-Spring-Damper (MSD) dynamic model that will be useful for the design of the manipulator and related vibration isolation system.

In this paper, first, the effect of simple impulsive force and torque disturbances on UAV dynamics is studied. Then, a coupled dynamic model of the UAV and a 1-DOF (Degree of Freedom) manipulator is developed and compared to a simplified decoupled model. Finally, a decoupled dynamic model is proposed for a 3-DOFs manipulator, and the simulation results for a real pick and place operation are presented and discussed.

The paper is organized as follows: Section II describes the simulated system, Section III introduces the UAV dynamics and control, Section IV and V describe the dynamic models and report the simulation results, and Section VI concludes the paper.

II. SIMULATED SYSTEM

The system under consideration is composed of an octocopter UAV (Fig. 1 and Table I) and a 3-DOFs robot manipulator (Fig. 2 and Table II). A 1-DOF manipulator (Table III) is used for comparison purposes in Section IV.

The UAV is a DJI S1000 heavy payload octocopter. It is composed of a carbon fiber reinforced plastics (CFRP) frame with eight arms carrying the motors and propellers, a flange for the battery that is also used as mechanical interface to the robot manipulator, and a landing gear including two retractable legs.

In this study, the frame of the UAV, the propellers, and the manipulator's links are considered as rigid bodies.

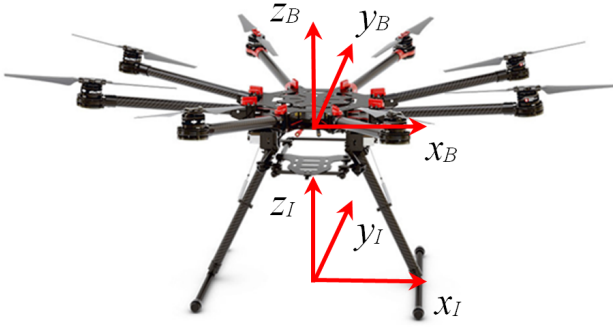


Fig. 1. UAV considered in this study with body (B) and inertial (I) reference frames

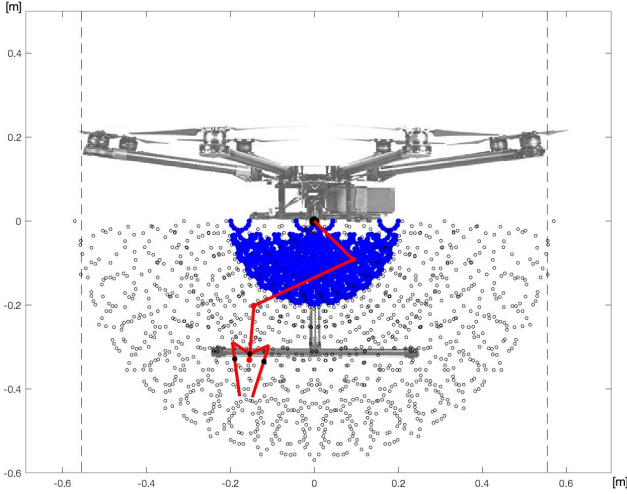


Fig. 2. UAV with 3-DOFs manipulator and its workspace (black circles: end-effector positions; blue circles: arm COG positions)

TABLE I. UAV INERTIAL PARAMETERS

Mass (kg)	Moment of Inertia around x_B (kgm^2)	Moment of Inertia around y_B (kgm^2)	Moment of Inertia around z_B (kgm^2)
4.2	0.4097	0.4097	0.3421

TABLE II. 3-DOFs MANIPULATOR GEOMETRICAL AND INERTIAL PARAMETERS

Link nr.	Mass (kg)	Length (m)	Moment of Inertia (Barycentric, kgm^2)	Center of Gravity (m)
1	0.529	0.13	$5.926 \cdot 10^{-4}$	0.114
2	0.657	0.26	$4.147 \cdot 10^{-3}$	0.209
3	0.129	0.13	$1.813 \cdot 10^{-4}$	0.065

TABLE III. 1-DOF MANIPULATOR GEOMETRICAL AND INERTIAL PARAMETERS

Link nr.	Mass (kg)	Length (m)	Moment of Inertia (Barycentric, kgm^2)	Center of Gravity (m)
1	1	0.13	$1.408 \cdot 10^{-3}$	0.065

III. UAV DYNAMICS AND CONTROL

A. Mathematical model

The octocopter UAV considered in the framework of this research is equipped with 8 identical propellers driven by electric motors. In order to minimize the gyroscopic effects, 4 propellers rotate clockwise, whereas the other 4 propellers rotate counter-clockwise. Lift and drag forces are assumed proportional to the square of the angular velocity of propellers. Ground effect is neglected. If all the elements of the UAV are assumed to be rigid, the system has 6 DOFs. Recent studies have shown that the first modes of vibration related to structural deformation of the octocopter appear above 40 Hz [12].

This UAV, as most of multi-rotor UAVs, is controlled acting on propellers angular velocities. Increasing or decreasing the 8 propellers angular velocities together changes the global lift force and generates the vertical motion. Increasing the angular velocity of propellers 1, 2, 3, 4 and decreasing of the same amount the angular velocities of propellers 5, 6, 7, 8 does not change the global lift force, but generates a torque about axis x_B that controls the roll rotation (ϕ), see Fig. 3. A similar operation carried out on two different groups of propellers (3, 4, 5, 6 and 7, 8, 1, 2) generates a torque about axis y_B that controls the pitch rotation (θ), see Fig. 3. Increasing the angular velocities of the 4 propellers rotating clockwise and decreasing of the same amount the angular velocity of the 4 propellers rotating counter-clockwise generates a torque about axis z_B that controls the yaw rotation (ψ).

It is worth highlighting that this UAV is under-actuated, because there is no possibility of directly generating thrusts in the x_B and y_B directions. Actually, the roll and pitch rotations alter the direction of the thrust with respect to the inertial reference frame and make possible motions along the x_I and y_I directions.

The three-dimensional (3D) dynamics of the UAV is described by Newton's equation (written in the inertial reference frame I) and by Euler's equation (written in the body reference frame B):

$$m_{UAV} \mathbf{a} = \mathbf{F}_g + [\mathbf{R}_{BI}] \sum_{i=1}^8 T_i \hat{\mathbf{e}}_{T_i}^B \quad (1)$$

$$[\mathbf{J}_{UAV}] \dot{\boldsymbol{\omega}} = -\boldsymbol{\omega} \times [\mathbf{J}_{UAV}] \boldsymbol{\omega} - \sum_{i=1}^8 \boldsymbol{\omega} \times [\mathbf{J}_r] \hat{\mathbf{e}}_{T_i}^B \Omega_i + \boldsymbol{\tau}$$

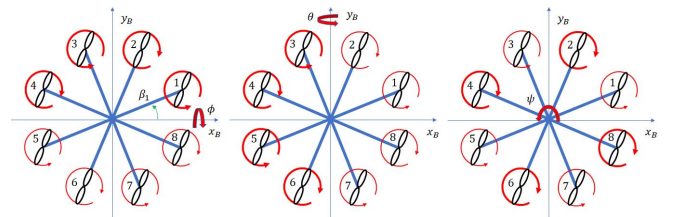


Fig. 3. UAV control, a thicker line means an increased velocity

In the Newton's equation in (1), m is the UAV mass, vector $\mathbf{a}$ the center of mass (CM) acceleration, $\mathbf{F}_g$ the gravity force, matrix $[R_B]$ the rotation matrix from body axes to inertial axes, which depends on Euler's angles. T_i is the thrust generated by the i^{th} propeller, and $\hat{\mathbf{e}}_{T_i}^B$ is the unit vector of this thrust force in the body reference frame.

In the Euler's equation in (1), $\boldsymbol{\omega}$ and $\dot{\boldsymbol{\omega}}$ are the angular velocity and acceleration of the UAV, $[J_{UAV}]$ is the inertia tensor of the UAV, which is assumed diagonal

$$[J_{UAV}] = \begin{bmatrix} I_x & 0 & 0 \\ 0 & I_y & 0 \\ 0 & 0 & I_z \end{bmatrix} \quad (2)$$

$[J_r]$ is the inertia tensor of the propellers, and Ω_i is the angular velocity of the i^{th} propeller. Finally, vector $\boldsymbol{\tau}$ contains the torques generated by the lift and drag forces about the body axes.

Equations (1) can be developed in scalar form making two important simplifications. The first simplification is related to UAV geometry: the unit vectors of propeller forces are assumed aligned to $\mathbf{z}_B$. The second simplification is related to the motion of the UAV. Since the focus of this research is on manipulation, which is performed when the UAV is in hovering flight, there is a small difference between the orientations of inertial axes and body axes, and the angular velocities in the body reference frame are almost equal to the derivatives of the Euler's angles.

Under these assumptions, the equations of motion of the UAV become:

$$\begin{aligned} m_{UAV} \ddot{x} &= \theta U_1 \\ m_{UAV} \ddot{y} &= -\phi U_1 \\ m_{UAV} \ddot{z} &= -m_{UAV} g + U_1 \\ I_x \ddot{\phi} &= \dot{\theta} \psi (I_y - I_z) - I_{zr} \dot{\theta} \Omega + U_2 \\ I_y \ddot{\theta} &= \dot{\phi} \psi (I_z - I_x) + I_{zr} \dot{\phi} \Omega - U_3 \\ I_z \ddot{\psi} &= \dot{\phi} \dot{\theta} (I_x - I_y) + U_4 \end{aligned} \quad (3)$$

with

$$\begin{aligned} \Omega &= (\Omega_1 - \Omega_2 + \Omega_3 - \Omega_4 + \Omega_5 - \Omega_6 + \Omega_7 - \Omega_8) \\ U_1 &= b(\Omega_1^2 + \Omega_2^2 + \Omega_3^2 + \Omega_4^2 + \Omega_5^2 + \Omega_6^2 + \Omega_7^2 + \Omega_8^2) \\ U_2 &= b[s(\beta_1)\Omega_1^2 + s(\beta_2)\Omega_2^2 + s(\beta_3)\Omega_3^2 + s(\beta_4)\Omega_4^2 \\ &\quad + s(\beta_5)\Omega_5^2 + s(\beta_6)\Omega_6^2 + s(\beta_7)\Omega_7^2 \\ &\quad + s(\beta_8)\Omega_8^2] \\ U_3 &= b[c(\beta_1)\Omega_1^2 + c(\beta_2)\Omega_2^2 + c(\beta_3)\Omega_3^2 + c(\beta_4)\Omega_4^2 \\ &\quad + c(\beta_5)\Omega_5^2 + c(\beta_6)\Omega_6^2 + c(\beta_7)\Omega_7^2 \\ &\quad + c(\beta_8)\Omega_8^2] \\ U_4 &= d(-\Omega_1^2 + \Omega_2^2 - \Omega_3^2 + \Omega_4^2 - \Omega_5^2 + \Omega_6^2 - \Omega_7^2 + \Omega_8^2) \end{aligned} \quad (4)$$

The term Ω is related to gyroscopic effects. It is equal to zero, if the angular velocities of the propellers rotating clockwise are equal to the angular velocities of the propellers rotating counter-clockwise. The term U_1 is the global thrust force that acts along $\mathbf{z}_B$. In hovering flight it is almost aligned to $\mathbf{z}_I$ and has small components along $\mathbf{x}_I$ and $\mathbf{y}_I$. The terms U_2, U_3, U_4 are the torques about the body axes generated by the propellers. The term I_{zr} is the barycentric moment of inertia of propellers around their z axis, and b

and d are the lift and drag coefficients of propellers. Angles β_i are the angles between $\mathbf{x}_B$ and the i^{th} rotor arm (see Fig. 3), and $s(\beta_i)$ and $c(\beta_i)$ are their sine and cosine.

A control system is needed to set the inputs U_1, U_2, U_3, U_4 in order to obtain the desired vertical position and attitude of the UAV. Many control strategies have been proposed to control the attitude of UAVs, see for example the review paper [13]. In this research, the control problem is simplified, since small oscillations about the hovering configuration are considered. In this case, the coupling terms between roll, pitch, and yaw equations become negligible and a set of independent PID controls [14] is used. The PID controllers are tuned by means of the Ziegler–Nichols method [15]. The equations of the controlled UAV have been implemented and solved in Simulink. Fig. 4 shows the flow-chart of the Simulink model.

B. Response to impulses

In order to study the main features of UAV dynamics, the 3D dynamic model with PID control is stimulated by triangular forces (or torque) impulses.

Fig. 5 shows that a roll torque impulse is compensated by the control system and, after the transient, the desired attitude is maintained. Since the variation in the roll angle changes the direction of the global thrust force U_1 , a transient lateral force along $\mathbf{y}_I$ appears. Since the system is under-actuated, the force along $\mathbf{y}_I$ causes a displacement of the UAV. It is worth noticing that the transient torque about the roll axis has no effect in $\mathbf{x}_I$ and $\mathbf{z}_I$ directions, and negligible effect on the pitch and yaw angles.

Similar effects are showed by a simulation with a transient pitch torque.

Simulations with impulsive yaw torque and impulsive vertical force show that the controller is able to cope with these disturbances with no effect in the other directions.

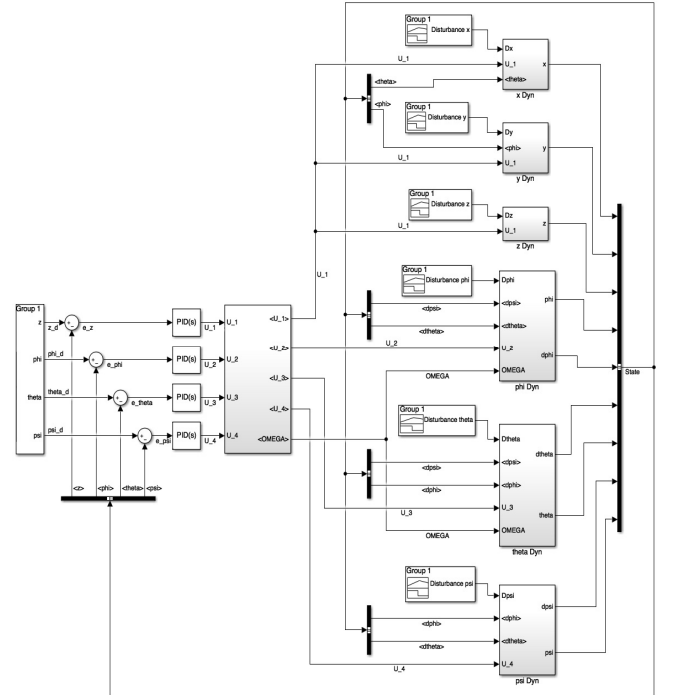


Fig. 4. Flow-chart of the Simulink model

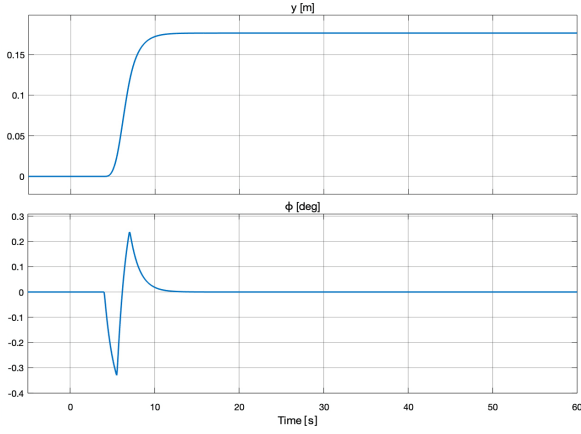


Fig. 5. Response to a triangular impulse about the roll axis

IV. COUPLED MODEL WITH 1-DOF MANIPULATOR

A. Mathematical model

The analysis presented in the previous section has shown that the dynamics of the roll, pitch, and yaw axes are essentially decoupled. Therefore, the development of a planar model in the vertical plane ($y_I z_I$) is meaningful. This model is developed to study the main features of the coupling between UAV dynamics and manipulator dynamics. A robot arm having 1 DOF is considered. Therefore, the model, which is represented in Fig. 6, has 4 DOFs, which are associated to the coordinates of the CM of the UAV (y and z), roll angle ϕ , and rotation θ of the robot arm with respect to the UAV. The equations of motion of the coupled system are derived with Lagrange's method.

The kinetic (E_k) and potential (E_p) energy of the system are given by the following equations:

$$E_k = \frac{1}{2} m_{UAV} (\dot{y}^2 + \dot{z}^2) + \frac{1}{2} I_g^{UAV} \dot{\phi}^2 + \frac{1}{2} m_l v_l^2 + \frac{1}{2} I_g^l (\dot{\phi} + \dot{\theta})^2 \quad (5)$$

$$E_p = m_{UAV} g z + m_l g z_{g,l}$$

in which I_g^{UAV} ($= I_x$) is the moment of inertia of the UAV about its CM, m_l and I_g^l are the mass and moment of inertia of the manipulator about its CM, v_l and $z_{g,l}$ are the velocity and vertical position of the robot CM, and g is the gravitational acceleration. If the position and velocity of the robot CM are expressed as a function of the selected coordinates, the following equations are obtained (see Fig. 6 for the definition of l and d):

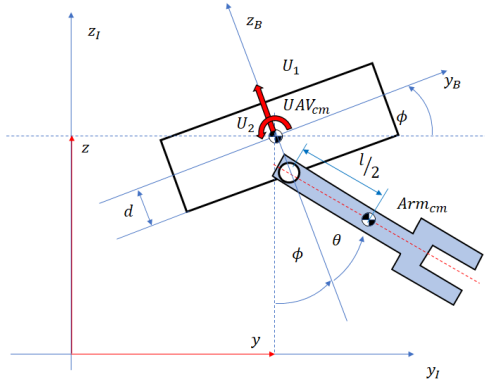


Fig. 6. Coupled model

$$E_k = \frac{1}{2} m_{UAV} \dot{y}^2 + \frac{1}{2} m_{UAV} \dot{z}^2 + \frac{1}{2} I_g^{UAV} \dot{\phi}^2 + \frac{1}{2} m_l \left\{ \dot{y}^2 + \dot{z}^2 + d^2 \dot{\phi}^2 + 2 \dot{y} d c(\phi) \dot{\phi} + 2 \dot{z} d s(\phi) \dot{\phi} + \frac{l^2}{4} \dot{\phi}^2 + \frac{l^2}{4} \dot{\theta}^2 + 2 + \frac{l^2}{4} \dot{\phi} \dot{\theta} + 2 \dot{y} \frac{l}{2} c(\phi + \theta) \dot{\phi} + 2 \dot{z} \frac{l}{2} s(\phi + \theta) \dot{\phi} + 2 \dot{y} \frac{l}{2} c(\phi + \theta) \dot{\theta} + 2 \dot{z} \frac{l}{2} s(\phi + \theta) \dot{\theta} + 2 d \frac{l}{2} c(\phi) \dot{\phi}^2 + 2 d \frac{l}{2} c(\phi) \dot{\phi} \dot{\theta} \right\} + \frac{1}{2} I_g^l \dot{\phi}^2 + \frac{1}{2} I_g^l \dot{\theta}^2 + \frac{1}{2} I_g^l \dot{\phi} \dot{\theta} \quad (6)$$

$$E_p = m_{UAV} g z + m_l g \left(z - d c(\phi) - \frac{l}{2} c(\phi + \theta) \right)$$

Lagrange's equations are:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = Q_i \quad (7)$$

In (7), $L = E_k - E_p$ is the Lagrange's function, q_i are the generalized coordinates (y , z , ϕ , and θ), and Q_i the generalized forces. The introduction of (6) in (7) leads to the dynamics equations of the coupled system:

$$(m_{UAV} + m_l) \ddot{y} + m_l \left(d c(\phi) + \frac{l}{2} c(\phi + \theta) \right) \ddot{\phi} + m_l \frac{l}{2} c(\phi + \theta) \ddot{\theta} + m_l \left\{ -\frac{l}{2} s(\phi + \theta) \dot{\theta}^2 + \left(d s(\phi) + \frac{l}{2} s(\phi + \theta) \right) \dot{\phi}^2 - 2 \frac{l}{2} s(\phi + \theta) \dot{\phi} \dot{\theta} \right\} = -s(\phi) U_1 \quad (8)$$

$$(m_{UAV} + m_l) \ddot{z} + m_l \left(d s(\phi) + \frac{l}{2} s(\phi + \theta) \right) \ddot{\phi} + m_l \frac{l}{2} s(\phi + \theta) \ddot{\theta} + m_l \left\{ +\frac{l}{2} c(\phi + \theta) \dot{\theta}^2 + \left(d c(\phi) + \frac{l}{2} c(\phi + \theta) \right) \dot{\phi}^2 - 2 \frac{l}{2} c(\phi + \theta) \dot{\phi} \dot{\theta} \right\} + (m_{UAV} + m_l) g = c(\phi) U_1 \quad (9)$$

$$m_l \left(d c(\phi) + \frac{l}{2} c(\phi + \theta) \right) \ddot{y} + m_l \left(d s(\phi) + \frac{l}{2} s(\phi + \theta) \right) \ddot{z} + \left[I_g^{UAV} + I_g^l + m_l \left(d^2 + \frac{l^2}{4} \right) + 2 d \frac{l}{2} \right] \ddot{\phi} + m_l d \frac{l}{2} c(\theta) \ddot{\theta} - m_l \left\{ 2 d c(\theta) \dot{z} \dot{\phi} + 2 d \frac{l}{2} s(\theta) \dot{\phi} \dot{\theta} + d \frac{l}{2} s(\theta) \dot{\phi}^2 \right\} + m_l g \left(d s(\theta) + \frac{l}{2} s(\phi + \theta) \right) = U_2 \quad (10)$$

$$m_l \frac{l}{2} c(\phi + \theta) \ddot{y} + m_l \frac{l}{2} c(\phi + \theta) \ddot{z} + \left(I_g^l + m_l \frac{l^2}{4} + d \frac{l}{2} c(\theta) \right) \ddot{\phi} + \left(I_g^l + m_l \frac{l^2}{4} \right) \ddot{\theta} + m_l \frac{l}{2} s(\theta) \dot{\phi}^2 + m_l g \frac{l}{2} s(\phi + \theta) = Q_\theta \quad (11)$$

It is worth noticing that these equations hold true even for large rotations of the UAV and of the manipulator. The generalized forces along the y and z coordinates are the

components of the global thrust force U_1 . The generalized force along ϕ is U_2 , whereas the generalized force Q_θ is the torque exerted by the motor of the manipulator's joint. Inputs U_1 and U_2 are controlled by the PID controllers mentioned in Section III. The PID parameters are equal to the ones adopted in Section III for axes z and ϕ .

In the next subsection, the effect of a prescribed motion of the manipulator on the position and attitude of the UAV is studied. Therefore, θ and its derivatives are prescribed and (11) becomes an algebraic equation that makes possible the calculation of Q_θ . Therefore, only differential equations (8), (9), and (10) have to be solved to analyze the dynamic behavior of the coupled system. This system of equations has been implemented and solved in Simulink.

B. Response of the coupled system

The simulated motion of the manipulator is a rotation from the vertical position (with the end-effector pointing downwards) to a maximum angular position ($\theta = 65.1$ deg), and back to the vertical position. A triangular velocity profile is prescribed with maximum velocity equal to 0.4545 rad/s and duration 10 s.

Fig. 7 (blue lines) shows the effect of manipulator motion on the UAV that is calculated using the developed coupled model. The reaction torque on the UAV, generated by the motion of the manipulator's joint, modifies the attitude of the UAV (angle ϕ), but the PID control counteracts this effect and in the end it guarantees the desired attitude ($\phi = 0$ deg for $t > 15$ s).

The transient variation of the attitude of the UAV changes the direction of the global thrust force U_1 with respect to the inertial reference frame. U_1 is an external force for the coupled dynamic system. There is a very small perturbation in the vertical position of the UAV with respect to the reference position, which disappears after the end of the manipulator motion, owing to the effect of the controller. The most important effect is a lateral displacement of the UAV, which does not extinguish at the end of the manipulator motion, and causes a final position error of about 0.06 m. This effect takes place because the inclination of the thrust force generates a lateral force component ($-s(\phi)U_1$), which is not compensated by the control system.

V. ROBOT DYNAMICS

A. Multibody Model of UAV and 3-DOFs Manipulator

In this subsection, a simplified multibody model of the UAV and a 3-DOFs planar manipulator is presented. This model synergistically uses Working Model 2D and Simulink. A pick and place operation is dealt with, in which the manipulator moves from the stowed configuration to reach an object located below the UAV (see Fig. 8). First, the force and torque reactions exerted by the manipulator on the UAV are computed using Working Model 2D. Then, the calculated force and torque reactions are used as an input to the Simulink UAV model in order to estimate UAV dynamics during the manipulation.

In the WM analysis, the UAV control is simplified assuming that (i) the manipulator base is free to slide along a horizontal guide (to simulate the fact that the UAV is not actuated along this direction), and (ii) the manipulator's base can not rotate (roll angle ϕ always equal to zero, to simulate an ideal attitude control with zero error).

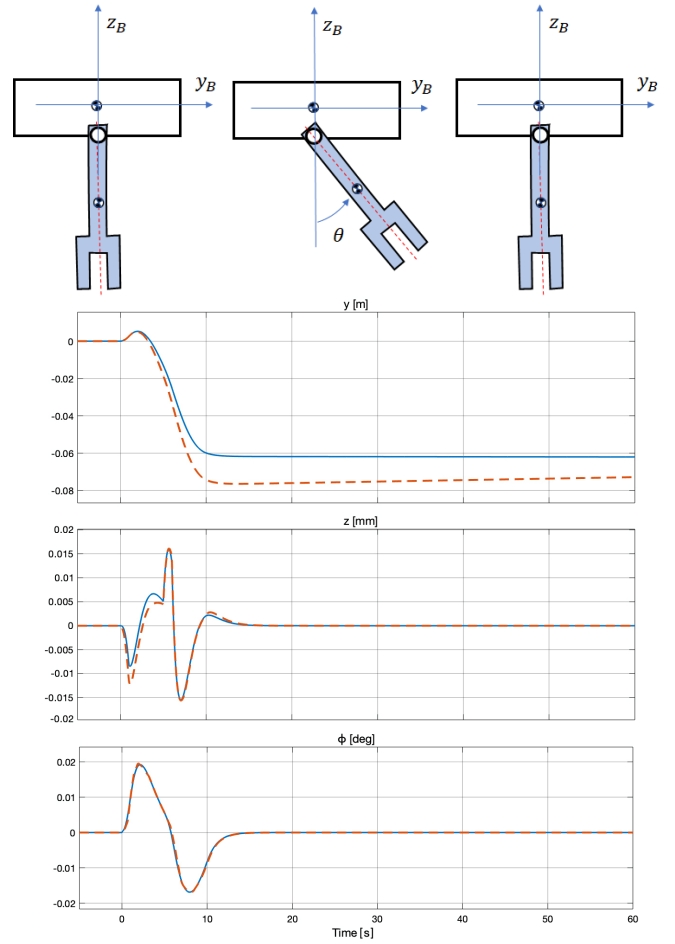


Fig. 7. Dynamic response of the coupled model (blue lines) and of the decoupled model (red dashed lines)

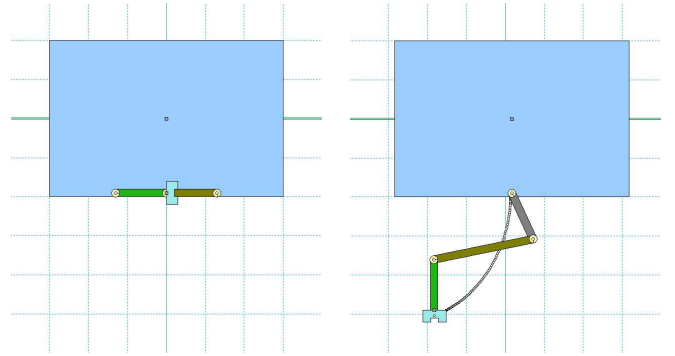


Fig. 8. 3-DOFs manipulator in stowed configuration (left) and in extended configuration (right)

UAV dynamics is simulated by means of the model of section III.

The proposed 2-step simulation procedure is an approximation (since it considers the manipulator and UAV as decoupled systems) of the actual system dynamics. In order to demonstrate that it is a reasonable approximation of system dynamics, the proposed approach is applied to a 1-DOF manipulator (see Table III for the manipulator geometrical and inertial parameters), and simulation results (red dashed lines) are compared to those of the coupled model (blue lines), see Fig. 7. It can be noticed that negligible errors are present in the variables z and ϕ , and that the presence of a y displacement of the system is

predicted by both models, even if an approx. 10% error is present when the decoupled model is used.

B. Simulation Results – UAV and 3-DOFs Manipulator Decoupled Model

The simulation results for the pick and place operation described in the previous subsection are presented here below. Fig. 9 shows the lateral displacement, the vertical displacement, and the roll angle of the UAV caused by the motion of the 3-DOFs manipulator.

It can be noticed that at the end of the operation the z and ϕ variables return to zero thanks to the UAV control system, whereas a net y displacement of about 0.11 m is generated, which cannot be compensated by the UAV control during hovering flight. Hence, owing to the displacement of the UAV, there is a position error between the end-effector and the object that has to be compensated.

VI. CONCLUSIONS

In this paper, first, the analysis of the dynamic response of a UAV to impulsive force/torque disturbances is carried out. In particular, it is showed that the system is able to suppress torque disturbances in the roll/yaw directions, but an undesired horizontal displacement is generated. Then, using a 1-DOF manipulator, it is demonstrated that a simplified decoupled model is effective in estimating the UAV-manipulator dynamics and, in particular, is able to predict (with an approx. 10% error) the lateral displacement of the system. This displacement is due to the fact that the

system cannot be actuated in the related direction while it is in hovering flight. Finally, the decoupled model is used to simulate the dynamics of a UAV equipped with a 3-DOFs planar manipulator during a pick and place operation. The simulation results show that in this scenario an undesired lateral displacement of the system is also present, which could jeopardize the manipulator precision.

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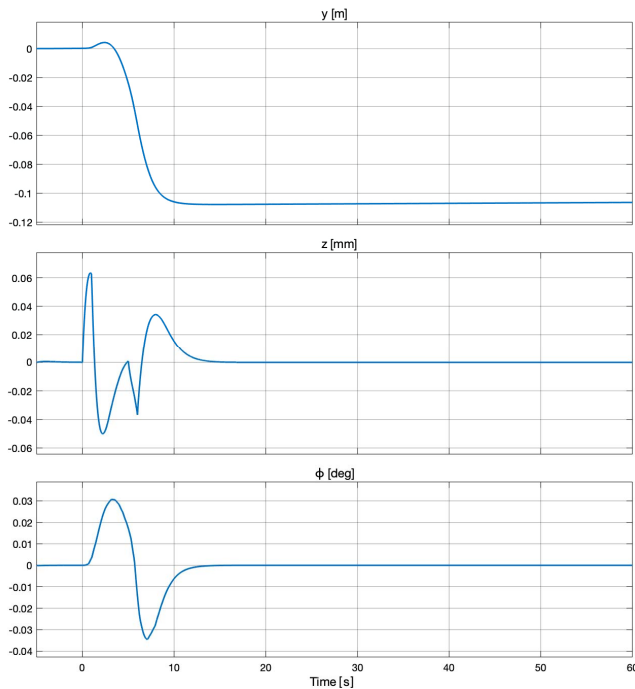


Fig. 9. Dynamic response of the decoupled model